

AI Game Asset Factory

Version 1.0

DATA STORAGE STRATEGY

SQLite Primary relational database.

JSON Portable metadata files.

Filesystem Binary asset storage.

ENTITY RELATIONSHIP DIAGRAM

Project 1 → N Assets 1 → N Animations 1 → 1 StyleProfile

Asset 1 → N Animations 1 → 1 Metadata

PROJECTS TABLE

projects Columns project_id TEXT PRIMARY KEY name TEXT NOT NULL description TEXT style_profile_id TEXT output_directory TEXT created_at DATETIME updated_at DATETIME

STYLE PROFILES TABLE

style_profiles Columns style_profile_id TEXT PRIMARY KEY name TEXT base_prompt TEXT negative_prompt TEXT palette_description TEXT cfg_scale REAL steps INTEGER sampler TEXT seed_strategy TEXT created_at DATETIME updated_at DATETIME

ASSETS TABLE

assets Columns asset_id TEXT PRIMARY KEY project_id TEXT asset_type TEXT name TEXT file_path TEXT thumbnail_path TEXT width INTEGER height INTEGER is_tileable BOOLEAN is_animated BOOLEAN created_at DATETIME updated_at DATETIME

Valid Asset Types character weapon prop building environment background tile effect ui

ANIMATIONS TABLE

animations Columns animation_id TEXT PRIMARY KEY project_id TEXT asset_id TEXT animation_type TEXT fps INTEGER frame_count INTEGER output_path TEXT created_at DATETIME

Valid Animation Types idle walk run jump attack hurt death

METADATA TABLE

metadata Columns metadata_id TEXT PRIMARY KEY asset_id TEXT json_path TEXT created_at DATETIME

GENERATION JOBS TABLE

generation_jobs Columns job_id TEXT PRIMARY KEY project_id TEXT job_type TEXT status TEXT started_at DATETIME completed_at DATETIME error_message TEXT

Valid Status pending running completed failed cancelled

MODEL REGISTRY TABLE

model_registry Columns model_id TEXT PRIMARY KEY name TEXT model_type TEXT version TEXT local_path TEXT is_active BOOLEAN created_at DATETIME

Valid Model Types generation animation classification background_removal upscaler

REFERENCE IMAGES TABLE

reference_images Columns reference_image_id TEXT PRIMARY KEY style_profile_id TEXT file_path TEXT weight REAL created_at DATETIME

JSON METADATA SCHEMA

Example { "asset_id": "uuid", "project_id": "uuid", "name": "tree_oak_001", "asset_type": "environment", "style_profile": "storybook_fantasy", "width": 512, "height": 512, "tileable": false, "animated": false, "tags": ["tree", "oak", "forest"], "generation_prompt": "", "created_at": "" }

FILESYSTEM STRUCTURE

projects/ |— fantasy_rpg/ |— project.json |— style_profile.json |— assets/ | |— characters/ | |— props/ | |— buildings/ | |— environments/ | |— tiles/ | |— effects/ |— animations/ |— metadata/ |— thumbnails/ |— exports/ |— cache/

NAMING CONVENTION

Characters character_<name>_<index> Example character_knight_001

Weapons weapon_<name>_<index> Example weapon_sword_001

Buildings building_<name>_<index> Example building_house_001

Animations <asset_name>_<animation_type> Example character_knight_001_walk

DATA INTEGRITY RULES

- Rule 1 Deleting a project deletes all child assets.
- Rule 2 Deleting an asset deletes child animations.
- Rule 3 Style profiles cannot be deleted while linked to projects.
- Rule 4 Every asset must belong to exactly one project.
- Rule 5 Every animation must reference an existing asset.
- Rule 6 All generated files must exist on disk before database commit.

BACKUP STRATEGY

Automatic Every project save.

Manual Export Project Archive Format ZIP Contents Assets Animations Metadata Style Profile Project Configuration